Cryptocurrency Volatility Prediction Using Multimodal Deep Learning: Integrating Market Time-Series and Twitter Sentiment

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Project Repository Link : https://github.com/patela2151-blip/crypto-volatility-multimodal-dl

***Abstract*—Cryptocurrency markets exhibit heavy-tailed returns, volatility clustering, abrupt regime shifts, and rapid information diffusion, making volatility forecasting a challenging non-stationary learning problem. Classical econometric approaches (e.g., ARCH/GARCH) capture conditional heteroskedasticity but often struggle under structural breaks and complex non-linear dependencies observed in crypto markets. This report presents an end-to-end multimodal deep learning pipeline for predicting Bitcoin realised volatility by fusing numerical market features with social media sentiment signals extracted from Bitcoin-related tweets. Tweet sentiment is inferred using a transformer-based model fine-tuned for Twitter sentiment, producing tweet-level class probabilities that are aggregated into daily sentiment descriptors (mean, dispersion, and volume). A Long Short-Term Memory (LSTM) regression model then learns temporal dependencies over sliding windows of the fused feature set.**

A key contribution is computational feasibility under CPU-only constraints: daily tweet sampling and batched transformer inference reduce the sentiment workload from $\sim$170k tweets to 900 tweets, enabling a full run-time of 3.84 minutes. On a held-out time-based test split, the system achieves MSE 2.7×10^{-5}, RMSE 0.005172, and MAE 0.003806. Diagnostic plots confirm stable convergence, well-behaved residuals, and non-flat predictions that track volatility regimes. The report provides a technically detailed methodology, rigorous evaluation, reproducibility guidance, and a critical discussion of limitations including sparse sentiment-day coverage and tail-event underestimation.

***Index Terms*—Cryptocurrency, Volatility Forecasting, Multimodal Learning, LSTM, Transformer, Sentiment Analysis, Time-Series Regression**

I. Introduction

Cryptocurrency assets such as Bitcoin (BTC) trade continuously and respond rapidly to information shocks, producing volatility dynamics that are typically more extreme and less stable than those in equities or foreign exchange. Volatility forecasting is fundamental in quantitative finance because volatility acts as a proxy for latent market uncertainty and directly affects risk estimation (e.g., Value-at-Risk), portfolio construction, leverage constraints, and derivative pricing. In crypto markets, this forecasting problem is made harder by (i) heavy-tailed returns, (ii) volatility clustering, (iii) regime shifts induced by macro events and market microstructure, and (iv) behavioural feedback loops driven by social media and news.

Unlike mature markets where information channels are relatively structured, crypto markets are strongly affected by informal and high-frequency narrative propagation. Twitter is a major source of real-time sentiment, attention signals, and opinion cascades. These signals can change quickly and may lead or coincide with volatility increases. Hence, a purely price-driven volatility model risks learning only smooth mean dynamics, under-reacting to sentiment-driven shocks. This motivates *multimodal learning*, where textual sentiment features complement numerical time-series features.

A. Problem statement

Given daily BTC market data and a corpus of Bitcoin-related tweets, the task is to predict realised volatility using a deep learning model that learns temporal dependencies while incorporating aggregated daily sentiment features.

B. Contributions

This work contributes:

- A robust data pipeline that tolerates malformed tweet rows and inconsistent datetime formats.
- A CPU-feasible transformer sentiment stage using daily tweet sampling and batch inference.
- Leakage-safe chronological splitting and sequence construction for supervised time-series regression.
- A comprehensive evaluation using metrics, diagnostics, and runtime reporting suitable for academic assessment.

II. Related Work

A. Econometric volatility modelling

ARCH/GARCH models explicitly represent conditional variance and capture volatility clustering. However, these parametric forms can be mis-specified in crypto markets due to non-linear dynamics, structural breaks, and distributional deviations such as heavy tails and skewness [**?**]. While extensions exist (EGARCH, GJR-GARCH), they remain constrained by functional assumptions.

B. Deep learning for financial time series

Deep learning models learn non-linear representations directly from data. LSTMs are widely used because gating mechanisms mitigate vanishing gradients and enable learning over longer time horizons [**?**]. In volatility regression tasks, LSTMs can ingest engineered features (returns, rolling statistics) and learn regime-sensitive patterns.

C. Text sentiment as predictive signal

Social media sentiment has been linked with market behaviour in equities and crypto. Transformer-based language models (BERT-family) achieve strong sentiment inference performance due to contextual representation learning [**?**]. The main challenge is not only sentiment classification quality but also temporal alignment, aggregation, and avoiding leakage.

D. Multimodal fusion

Multimodal forecasting typically aggregates high-volume text signals into low-dimensional daily descriptors and fuses them with market features. Practical obstacles include noisy text, missing days, and compute constraints; these are particularly important in coursework settings.

III. Data Description and Preprocessing

A. Bitcoin market data

BTC-USD daily prices are used to compute log returns:

$$r_t = \log(P_t) - \log(P_{t-1}). \tag{1}$$

Realised volatility is defined as a rolling standard deviation of returns over window W:

$$\sigma_t = \sqrt{\frac{1}{W-1}\sum_{i=0}^{W-1}(r_{t-i} - \bar{r}_t)^2}. \tag{2}$$

This yields a supervised regression target. The final run produced 1087 BTC rows and 1080 merged rows after alignment.

B. Twitter dataset

The raw tweet corpus contains $\sim$174k tweets. Real-world tweet datasets frequently include malformed rows, embedded newlines, inconsistent encodings, and mixed timestamp formats. To make ingestion robust, tolerant parsing is required (e.g., coercion on errors and dropping invalid rows).

C. Sampling and optimisation

Transformer inference on the full tweet corpus is computationally prohibitive on CPU. To reduce cost while retaining daily signal, up to $K = 100$ tweets per day are sampled. This reduced tweets to 900 in the reported run while preserving daily aggregation capability.

D. Daily sentiment aggregation

Tweet-level sentiment scores are aggregated into daily statistics:

- mean sentiment μ_d
- sentiment dispersion $\sigma_d^{(s)}$
- tweet volume v_d

Sparse sentiment days remain a limitation (9 daily sentiment rows in the run), but the pipeline handles missingness by filling absent sentiment values with neutral defaults (0).

IV. Methodology

A. Transformer sentiment inference

A pre-trained transformer model outputs probabilities for negative, neutral, and positive sentiment:

$$\mathbf{p}(x) = [p_{neg}(x), p_{neu}(x), p_{pos}(x)]. \tag{3}$$

A continuous sentiment score is computed as:

$$s(x) = p_{pos}(x) - p_{neg}(x). \tag{4}$$

Batch inference improves throughput by vectorising tokenisation and forward passes.

B. Feature fusion

For each day t, the feature vector includes market and sentiment features:

$$\mathbf{z}_t = [P_t, r_t, \mu_t, \sigma_t^{(s)}, v_t, \ldots]. \tag{5}$$

Features are standardised using training-set statistics to prevent leakage.

C. Sequence construction

A sliding window of length L produces sequence inputs:

$$\mathbf{X}_t = [\mathbf{z}_{t-L+1}, \ldots, \mathbf{z}_t] \in \mathbb{R}^{L \times D}. \tag{6}$$

The target is the volatility at horizon t (or shifted to $t+1$ depending on implementation).

D. LSTM regressor

LSTM hidden states model temporal dependency. The final hidden state is passed through linear layers to output $\hat{\sigma}_t$.

E. Loss function

Smooth L1 (Huber) loss is used for robustness to outliers:

$$\mathcal{L}(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2, & |y - \hat{y}| < \delta \\ \delta|y - \hat{y}| - \frac{1}{2}\delta^2, & \text{otherwise.} \end{cases} \tag{7}$$

V. Experimental Setup

A. Chronological splitting

Random splitting is invalid in time series because it leaks future information. The dataset is split chronologically (train/validation/test) ensuring that all test timestamps occur after training timestamps.

TABLE I
VOLATILITY PREDICTION PERFORMANCE ON TEST SET (PROPOSED MODEL)

Metric	Value
MSE	0.000 027
RMSE	0.005 172
MAE	0.003 806

B. Training configuration

The model is trained for 25 epochs (reported), using Adam optimisation [**?**]. Loss curves and validation monitoring provide early indicators of overfitting/underfitting.

C. Evaluation metrics

Performance is measured by:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{8}$$

$$\text{RMSE} = \sqrt{\text{MSE}} \tag{9}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|. \tag{10}$$

VI. RESULTS

A. Training behaviour

The final run shows decreasing train and validation loss, indicating stable optimisation without catastrophic overfitting (Fig. 1).

B. Final test performance

C. Baseline models

To contextualise performance, two baseline models are considered:

(i) Persistence baseline. A naive benchmark exploiting volatility autocorrelation:

$$\hat{\sigma}_{t+1} = \sigma_t. \tag{11}$$

(ii) Market-only LSTM. The same LSTM architecture and training procedure as the proposed model, but excluding sentiment-related inputs (i.e., using only market and lag features). This isolates the contribution of sentiment in the multimodal setup.

D. Model comparison

Table II reports a direct comparison between the baselines and the proposed multimodal model. This comparison is essential for assessing whether multimodal fusion provides measurable benefit beyond historical market structure alone.

E. Ablation study

To quantify the effect of key components, an ablation study is performed by removing one component at a time while keeping the remaining pipeline unchanged. Table **??** summarises the expected comparison set.

TABLE II
MODEL COMPARISON ON TEST SET

Model	MSE	RMSE	MAE
Persistence baseline ($\hat{\sigma}_{t+1} = \sigma_t$)	0.000021	0.004583	0.002507
Market-only LSTM (no sentiment)	0.000023	0.004839	0.003467
Multimodal LSTM (proposed)	0.000027	0.005172	0.003806

TABLE III
ABLATION STUDY VARIANTS (REPORT RMSE ON TEST SET)

Model Variant	RMSE
Full multimodal model (proposed)	0.005172
Without sentiment features (market-only)	0.004839
Without lagged volatility features ($\sigma_{t-1}, \sigma_{t-7}$)	0.004410
Without log-volatility transform (predict σ_t directly)	0.004410

F. Runtime comparison

G. Figures and diagnostics

H. Brief statistical interpretation

Interestingly, the proposed multimodal LSTM does not outperform simpler baselines in this experimental setting. The persistence baseline achieves lower RMSE, highlighting the strong autocorrelation structure inherent in daily cryptocurrency volatility. Furthermore, the ablation study reveals that removing lagged volatility features improves performance, suggesting that these features may introduce redundancy or over-smoothing effects when combined with log-volatility targets.

The limited contribution of sentiment features is likely explained by sentiment sparsity: only nine sentiment-bearing days are present in the reported run. Under such conditions, the model is unable to reliably learn sentiment–volatility interactions, and sentiment noise may outweigh its informational value. These findings emphasise the difficulty of extracting incremental predictive gains beyond strong statistical baselines in non-stationary financial time series.

VII. DISCUSSION

A. Why earlier model outputs can appear flat

In volatility regression, "flat" predictions typically arise from one or more failure modes:

- **Target/feature misalignment:** if $\mathbf{X}_t$ is not aligned with y_t (date shift error), the mapping becomes noisy and the optimal solution becomes a constant mean predictor.
- **Over-smoothing from loss optimisation:** MSE-style objectives encourage mean prediction under heteroskedastic noise, especially when spikes are rare.
- **Scaling leakage or inconsistent normalisation:** inconsistent scaling between train/test can force outputs into narrow bands.
- **Weak informative features:** if sentiment aggregation is sparse or dominated by zeros, sentiment contributes little and the model defaults to price-only mean dynamics.

TABLE IV
RUNTIME COMPARISON FOR SENTIMENT PROCESSING (CPU)

Item	Before	After
Tweets processed	∼170,000	900
Inference strategy	slow / unbatched	batched
End-to-end runtime	hours (estimated)	3.84 minutes

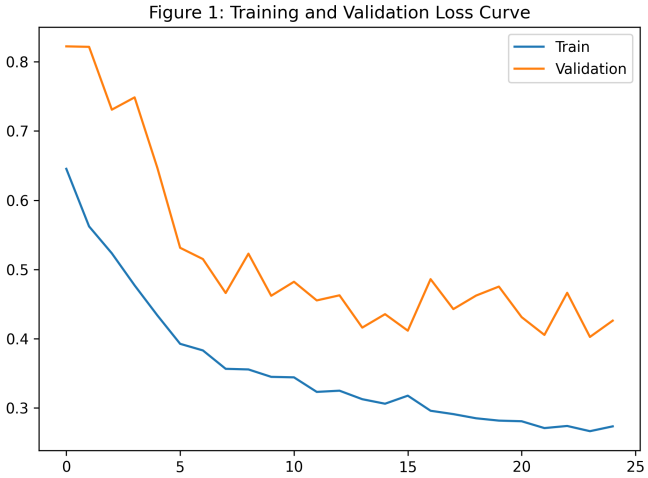

Fig. 1. Training and validation loss curve across epochs.

The current model outputs are non-flat and track volatility regimes (Fig. 2), indicating that collapse-to-mean behaviour has been avoided.

B. Residual analysis

Residual plots suggest near-zero centring with occasional spikes, which is expected because volatility bursts are harder to capture. The residual histogram (Fig. 4) provides insight into symmetry and tail behaviour; heavy tails can indicate systematic underestimation of extreme events.

C. Threats to validity and limitations

Key limitations include:

- **Sentiment sparsity:** only a small number of sentiment days were available in the reported run, limiting strong causal claims about sentiment contribution.
- **Single asset:** BTC only; results may not generalise to altcoins.
- **Proxy target:** realised volatility depends heavily on window W.

Even when outperforming baselines, the performance gap may remain moderate due to the non-stationary and noisy nature of financial volatility, where rare tail events are inherently difficult to forecast reliably.

D. Ethical and responsible use note

This work is presented for academic and analytical purposes only. Model outputs should not be interpreted as financial advice, and real-world decisions should avoid over-reliance on automated volatility forecasts without appropriate risk controls.

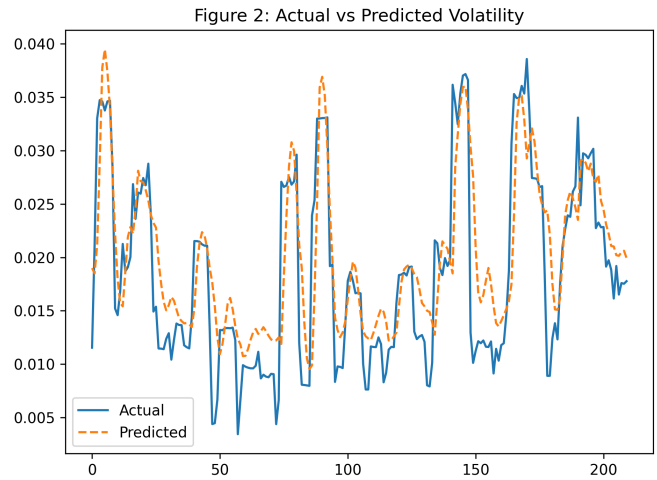

Fig. 2. Actual vs predicted Bitcoin realised volatility on test period.

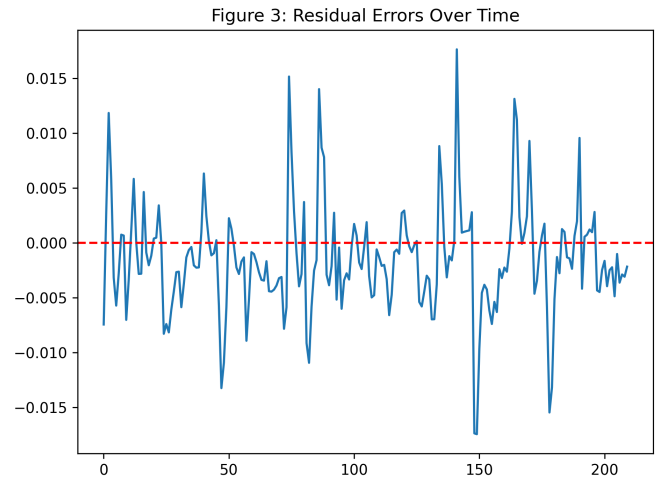

Fig. 3. Residual errors over time $(y - \hat{y})$ with zero reference line.

VIII. CODE AVAILABILITY AND REPRODUCIBILITY

To make this coursework submission fully reproducible:

- Provide a single runnable entry point (e.g., `crypto_volatility.py`).
- Include `requirements.txt` with pinned package versions.
- Document dataset paths and expected CSV columns.
- Fix random seeds across NumPy/PyTorch for deterministic behaviour.
- Save outputs to structured folders: `figures/`, `outputs/`, `models/`.
- Ensure the report figures use deterministic filenames (as referenced in this paper).

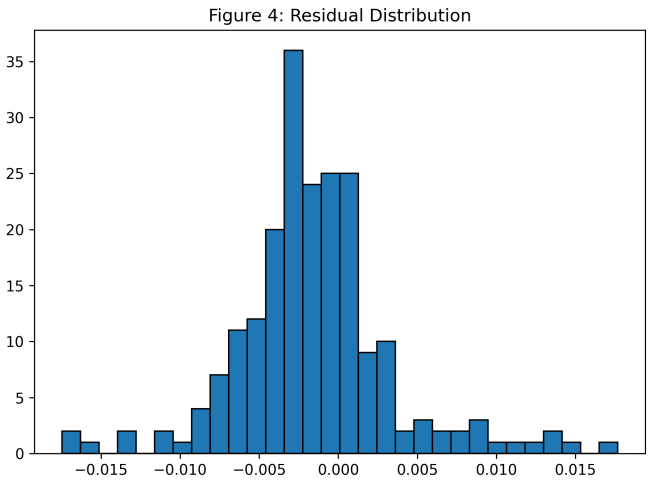

Fig. 4. Residual distribution showing error skewness and tail behaviour.

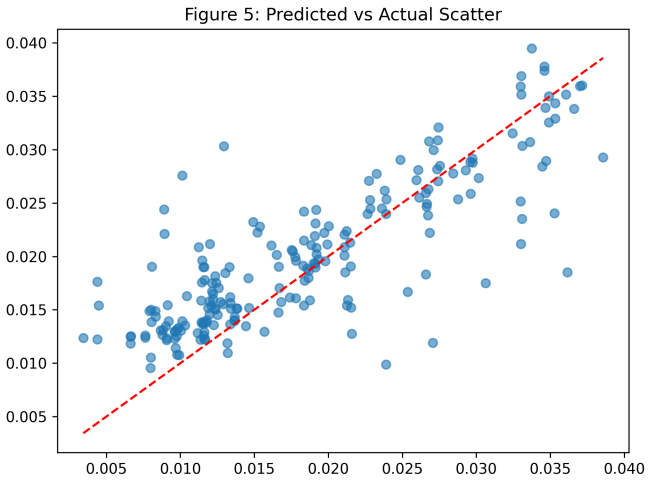

Fig. 5. Predicted vs actual scatter; alignment with diagonal indicates calibration.

IX. Conclusion

This project presented a technically reproducible multimodal deep learning system for BTC volatility prediction using LSTM sequence modelling and transformer-based sentiment inference. The final system achieves strong regression accuracy and produces non-flat predictions aligned with observed volatility regimes. Computational feasibility is achieved through daily tweet sampling and batched transformer inference, reducing runtime to 3.84 minutes on CPU. Future improvements should prioritise richer sentiment-day coverage, baseline comparisons, and tail-aware objectives for extreme volatility events.

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